# A Counterexample to Necessity of the Logarithmic $\overline{T}_N$ Condition

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## 1 Source Question

The source paper is

> P. M. Berna, O. Blasco, G. Garrigos, E. Hernandez, and T. Oikhberg, *Embeddings and Lebesgue-type inequalities for the greedy algorithm in Banach spaces*, arXiv:1707.07513.

In Section 7 the paper asks:

> **Question 2.** Characterize the systems for which
>
> $$\max\{L_N, L_N^*\} \lesssim \log(N+1).$$
>
> Concerning Question 2,
>
> $$\overline{T}_N(D, D^*) \lesssim \log(N+1)$$
>
> gives a sufficient condition, but the authors do not know whether it is necessary.

The source passage is shown in Figure 1.

## 2 Result

**Theorem 1.** *There is a seminormalized complete biorthogonal system $(\mathcal{B}, \mathcal{B}^*)$ in a Banach space $X$ such that*

$$\max\{L_N(\mathcal{B}, X), L_N(\mathcal{B}^*, \widehat{X})\} \le C\log(N+1), \qquad N \ge 2,$$

*but*

$$\overline{T}_N(D, D^*) \ge c(\log N)^2, \qquad N \ge 2.$$

*Consequently, the sufficient condition $\overline{T}_N(D, D^*) \lesssim \log(N+1)$ is not necessary for the logarithmic Lebesgue estimate in Question 2.*

## 3 Idea of the Proof

The construction puts two Lorentz sequence spaces next to each other. One has fundamental function $\eta(n) \simeq \sqrt{n\log n}$, and the other has the reciprocal fundamental function $\xi(n) \simeq \sqrt{n/\log n}$, so that $\eta(n)\xi(n) = n$. The $\ell_\infty$-direct sum makes the lower democracy function small, of size $\xi$, but leaves the upper democracy function large, of size $\eta$. The same happens in the dual because the two fundamental functions are reciprocal. Thus both primal and dual democracy ratios are only logarithmic, and unconditionality gives $L_N, L_N^* = O(\log N)$.

in particular all greedy bases.

**Corollary 7.6.** *Let $\{\mathbf{e}_n, \mathbf{e}_n^*\}_{n=1}^\infty$ be a complete s-biorthogonal system in $\mathbb{X}$. If $\mathcal{B} = \{\mathbf{e}_n\}_{n=1}^\infty$ is almost greedy, or $\{\mathbf{e}_n, \mathbf{e}_n^*\}_{n=1}^\infty$ is bidemocratic, then*

$$\max\left\{K_N, \widetilde{\mathbf{L}}_N, \mathbf{L}_N, g_N^*, \mu_N^*, \widetilde{\mathbf{L}}_N^*, \mathbf{L}_N^*\right\} \lesssim \overline{T}_N(D, D^*) \lesssim \ln(N+1)\,. \tag{7.10}$$

*Proof.* This follows from (7.6), using $\mu_N \approx 1$. □

We pose two questions.

**Question 1:** Characterize the systems $\{\mathbf{e}_n, \mathbf{e}_n^*\}$ for which $\overline{T}_N(D, D^*) \lesssim \mathbf{L}_N \ln(N+1)$.

**Question 2:** Characterize the systems for which $\max\{\mathbf{L}_N, \mathbf{L}_N^*\} \lesssim \ln(N+1)$.

Concerning Question 1, all the examples we have tested seem to satisfy this property. Concerning Question 2, $\overline{T}_N(D, D^*) \lesssim \ln(N+1)$ gives a sufficient condition, but we do not know whether it is necessary.

8. Examples

Figure 1: Source passage from arXiv:1707.07513, p. 20.

On the other hand $D(N)$ and $D^*(N)$ both dominate $\eta(N)$. Since

$$\sum_{j\le N} \frac{\eta(j)}{j}\Delta\eta(j) \asymp \sum_{j\le N} \frac{\log j}{j} \asymp (\log N)^2,$$

the $\overline{T}_N$-quantity is quadratically logarithmic. This separates the sufficient condition from the desired conclusion.

# 4 Construction

Set, for $n \ge 1$,

$$\lambda_n = 2 + \log n, \qquad \eta(n) = \sqrt{n\lambda_n}, \qquad \xi(n) = \sqrt{\frac{n}{\lambda_n}},$$

and put $\eta(0) = \xi(0) = 0$. The functions $x \mapsto \sqrt{x(2+\log x)}$ and $x \mapsto \sqrt{x/(2+\log x)}$ are increasing and concave on $[1, \infty)$, hence the sequences $\eta$ and $\xi$ are concave weights. They satisfy

$$\eta(n)\xi(n) = n, \qquad \frac{\eta(n)}{\xi(n)} = \lambda_n.$$

For a concave weight $\omega$, let $d(\omega)$ denote the discrete Lorentz space $\ell^1_\omega$ used in the source paper:

$$\|a\|_{d(\omega)} = \sum_{j=1}^{\infty} a_j^*\,\Delta\omega(j),$$

where $a^*$ is the nonincreasing rearrangement. The canonical unit-vector basis is 1-symmetric and has fundamental function exactly $\omega$:

$$\|\mathbf{1}_A\|_{d(\omega)} = \omega(|A|).$$

By the duality theorem for these Lorentz spaces recorded in the source paper, the closed coordinate span of the dual is the Marcinkiewicz space $m(\omega')$, where $\omega'(n) = n/\omega(n)$. A direct calculation gives

$$\|\mathbf{1}_A\|_{m(\omega')} = \omega'(|A|).$$

Now define

$$X = d(\eta) \oplus_\infty d(\xi),$$

and let $\mathcal{B}$ be the concatenated coordinate basis. This is a seminormalized complete 1-unconditional basis. Its closed dual coordinate span is

$$\widehat{X} = d(\eta)^* \oplus_1 d(\xi)^*.$$

# 5 Democracy Estimates

If $A$ has $k$ coordinates in the $d(\eta)$-summand and $\ell$ coordinates in the $d(\xi)$-summand, then

$$\|\mathbf{1}_A\|_X = \max\{\eta(k), \xi(\ell)\},$$

with the convention that $\eta(0) = \xi(0) = 0$. Since $\eta \geq \xi$, the upper democracy function is exactly

$$D(N) = \eta(N).$$

The lower democracy function satisfies

$$d(N) = \inf_{k+\ell=N} \max\{\eta(k), \xi(\ell)\} \asymp \xi(N).$$

Indeed, the upper bound is obtained by taking all $N$ coordinates from the $d(\xi)$-summand. For the lower bound, either $k \geq N/2$, when $\eta(k) \gtrsim \eta(N) \geq \xi(N)$, or $\ell \geq N/2$, when $\xi(\ell) \gtrsim \xi(N)$.

In the dual coordinate span, the same split gives

$$\|\mathbf{1}_A^*\|_{\widehat{X}} = \xi(k) + \eta(\ell),$$

because $\eta' = \xi$ and $\xi' = \eta$. Hence

$$D^*(N) = \sup_{k+\ell=N} \{\xi(k) + \eta(\ell)\} \asymp \eta(N),$$

and

$$d^*(N) = \inf_{k+\ell=N} \{\xi(k) + \eta(\ell)\} \asymp \xi(N).$$

Consequently the primal and dual democracy ratios obey

$$\mu_N \asymp \frac{D(N)}{d(N)} \lesssim \lambda_N \lesssim \log(N+1), \qquad \mu_N^* \asymp \frac{D^*(N)}{d^*(N)} \lesssim \log(N+1).$$

# 6 Lebesgue Constants

We use the following standard Konyagin–Temlyakov estimate, included for completeness in the 1-unconditional case.

**Lemma 1.** *Let $(e_n)$ be a 1-unconditional basis with superdemocracy ratio*

$$\mu_N = \sup_{\substack{|A|=|B|\le N \\ \varepsilon,\delta}} \frac{\|\mathbf{1}_{\varepsilon A}\|}{\|\mathbf{1}_{\delta B}\|}.$$

*Then $L_N \le 1 + \mu_N$.*

*Proof.* Let $\Gamma$ be a greedy set of size $N$ for $x$, and let $z$ be supported on a set $B$ with $|B| \le N$. Put

$$A = B \setminus \Gamma, \qquad C = \Gamma \setminus B.$$

Then $|A| \le |C|$. Since

$$x - P_\Gamma x = P_{(B\cup\Gamma)^c}(x - z) + P_A x,$$

1-unconditionality gives

$$\|P_{(B\cup\Gamma)^c}(x - z)\| \le \|x - z\|.$$

Let $\alpha = \min_{n\in\Gamma} |e_n^*(x)|$. The greedy choice gives $|e_n^*(x)| \le \alpha$ for $n \in A$, while $|e_n^*(x-z)| = |e_n^*(x)| \ge \alpha$ for $n \in C$. By lattice monotonicity and democracy, choosing $C_0 \subset C$ with $|C_0| = |A|$ when $A \ne \emptyset$,

$$\|P_A x\| \le \alpha\|\mathbf{1}_A\| \le \mu_N \alpha \|\mathbf{1}_{C_0}\| \le \mu_N \alpha \|\mathbf{1}_C\| \le \mu_N \|P_C(x - z)\| \le \mu_N \|x - z\|.$$

Taking the infimum over all such $z$ proves the claim. $\square$

Applying Lemma 1 to $\mathcal{B}$ and to its dual coordinate basis gives

$$L_N \lesssim \log(N + 1), \qquad L_N^* \lesssim \log(N + 1).$$

# 7 The $\overline{T}_N$ Lower Bound

Let $H(N) = D^*(N)$. From the previous section,

$$D(N) = \eta(N), \qquad H(N) \ge \eta(N).$$

Also $w_j = \eta(j)/j = \sqrt{\lambda_j/j}$ is decreasing. Abel summation therefore yields

$$T_N(D, D^*) = \sum_{j=1}^{N} w_j\, \Delta H(j) \ge \sum_{j=1}^{N} w_j\, \Delta\eta(j) = T_N(\eta, \eta).$$

The other order is even more immediate:

$$T_N(D^*, D) = \sum_{j=1}^{N} \frac{H(j)}{j} \Delta\eta(j) \ge T_N(\eta, \eta).$$

Thus $\overline{T}_N(D, D^*) \ge T_N(\eta, \eta)$.

It remains to estimate $T_N(\eta, \eta)$. Since

$$\eta(j)^2 = j(2 + \log j),$$

the discrete difference satisfies

$$\Delta(\eta^2)(j) \asymp 2 + \log j, \qquad j \ge 2.$$

Moreover

$$\frac{1}{2}\Delta(\eta^2)(j) \le \eta(j)\Delta\eta(j) \le \Delta(\eta^2)(j).$$

Consequently

$$T_N(\eta,\eta) = \sum_{j=1}^{N} \frac{\eta(j)}{j}\Delta\eta(j) \asymp \sum_{j=2}^{N} \frac{2+\log j}{j} \asymp (\log N)^2.$$

This proves Theorem 1.

# 8 Verification and Novelty Notes

The construction is deterministic and uses no numerical computation. The source-paper ingredients used above are the definitions of $D, D^*, T_N, \overline{T}_N$, the discrete Lorentz spaces $\ell^1_{\widehat{\omega}}$, and the duality $(\ell^1_{\widehat{\omega}})^* = m(\omega')$ for doubling weights with $\omega(n)/n \to 0$.

Cheap run indexes were searched for arXiv:1707.07513, the title, greedy Lebesgue constants, $T_N(D,D^*)$, $\overline{T}_N(D,D^*)$, and democracy keywords. The only nearby indexed results concern different greedy-basis questions. External web/arXiv searches for exact $\overline{T}_N(D,D^*)$ phrases and the source question found related greedy-basis literature but no answer to this necessity question. The novelty confidence is therefore medium-high, subject to expert review of nearby greedy-basis folklore.

# 9 Limitations and Human Review

This is not a characterization of all systems satisfying Question 2. It only shows that the sufficient condition $\overline{T}_N(D,D^*) \lesssim \log(N+1)$ is not necessary. It also does not address Question 1: the present example still has $\overline{T}_N(D,D^*) \lesssim L_N \log(N+1)$, so it is compatible with the authors' observed behavior for Question 1.

Recommended human checks:

- verify the dual coordinate-span calculation for the two Lorentz spaces and the $\oplus_\infty/\oplus_1$ direct sum;
- verify Lemma 1 under the exact convention for $L_N$ used in the source paper;
- check that no later paper has already recorded this direct-sum counterexample to the necessity of the $\overline{T}_N$-condition.

# References


[1] P. M. Berna, O. Blasco, G. Garrigos, E. Hernandez, and T. Oikhberg, *Embeddings and Lebesgue-type inequalities for the greedy algorithm in Banach spaces*, arXiv:1707.07513.

[2] S. V. Konyagin and V. N. Temlyakov, *A remark on greedy approximation in Banach spaces*, East J. Approx. 5 (1999), 365–379.